

Net acceptance summary

Four reportable issues · one open question · cross-tool coverage closed

Modelling decisions are defensible

Structure · geometry · ion identity (Mg²⁺ excluded) · water placement (143 / 146 well-supported) · disulfide bonds.

1SAR refinement is a real and substantial improvement over the deposited starting model.

1 R-free under-stated

≈ 0.012 below 4-tool consensus

Four code paths bracket R-free
0.211–0.221. Success-criterion
gap < 0.05 fails 3 of 4.

2 Round 7 ~~Round 6~~

0.212 vs 0.207 at oracle scaling

NCS-torsion-restraints
justification doesn't
reproduce.

3 Asn A 39 outlier

3-tool confirmed

Agent's 0.00% Ramachandran
outliers headline rounds 1
outlier to zero.

4 Peak inventory

5 peaks > 5σ not reported

23 + / 8 – peaks at 4σ; agent
listed 9. Max +6.18σ.

Open

Open question

K⁺ alternative for both ions

Best raw bond-length fit, but coordination number low and the protein is a known Ca²⁺ enzyme.

Needs crystallisation-buffer information.

Strengthened in this re-run (2026-05-05)

Servalcat sigmaa direct R-work/R-free → 4-tool R-free bracket

PHENIX 0.211 · gemmi 0.217 · REFMAC 0.213 · Servalcat 0.221 — agent 0.199 below all four.

Every 2026-05-04 finding reproduces deterministically against the same artefact.

Cross-tool coverage matrix fully green; T13 + T03 both have ≥ 4 oracles.